

SOFTWARE TESTING & QUALITY ASSURANCE

Install Visual Studio Code

Verification command code --version

Download from official website

Install recommended extensions: ES6, Jest, GitLens

Configure basic settings (theme, font size, etc.)

Install Node.js (Latest LTS version)

Verification commands node --version

npm --version

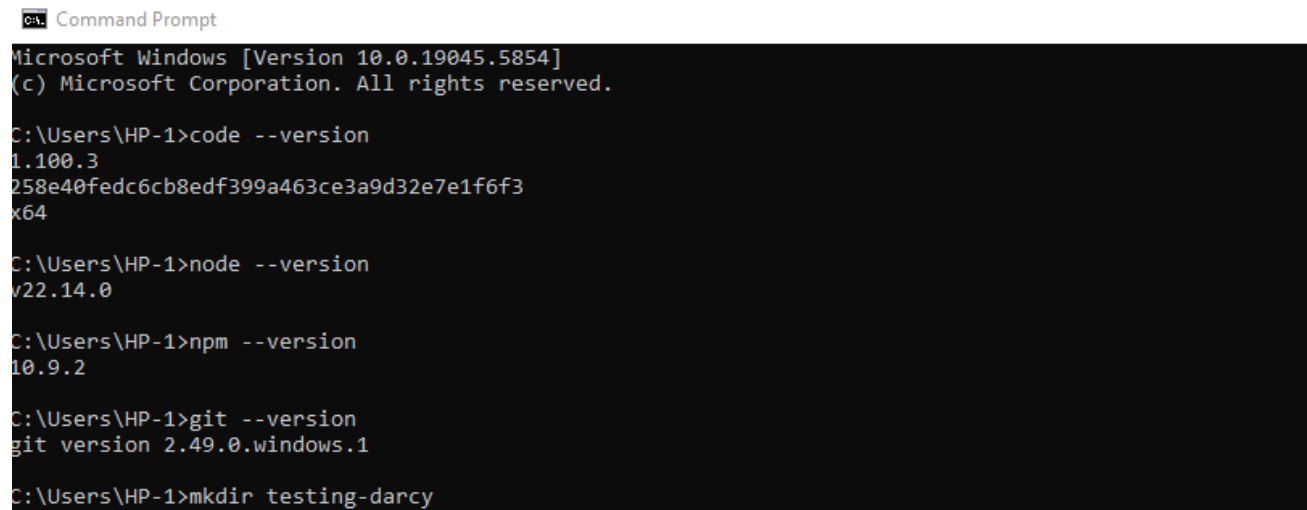
Download from nodejs.org
Verify npm is included
Test with simple JavaScript execution

Step 2: Version Control

Install Git

Verification and setup git --version
git config --global user.name "Your Name"
git config --global user.email "your.email@unilak.ac.rw"

Configure global settings
Generate SSH key for GitHub (optional)
Test basic git commands



```
cmd Command Prompt
Microsoft Windows [Version 10.0.19045.5854]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP-1>code --version
1.100.3
258e40fedc6cb8edf399a463ce3a9d32e7e1f6f3
x64

C:\Users\HP-1>node --version
v22.14.0

C:\Users\HP-1>npm --version
10.9.2

C:\Users\HP-1>git --version
git version 2.49.0.windows.1

C:\Users\HP-1>mkdir testing-darcy
```

Step 3: Initialize Test Project

Create project structure mkdir testing-your-name cd testing-your-name

C:\> Administrator: Command Prompt

```
C:\Windows\system32>cd .\testing-darcy\  
  
C:\Windows\System32\testing-darcy>npm init -y  
Wrote to C:\Windows\System32\testing-darcy\package.json:  
  
{  
  "name": "testing-darcy",  
  "version": "1.0.0",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \"Error: no test specified\" && exit 1"  
  },  
  "keywords": [],  
  "author": "",  
  "license": "ISC",  
  "description": ""  
}
```

Step 4: Install Testing Dependencies

Core testing frameworks npm install --save-dev jest
npm install --save-dev mocha chai npm install --save-dev
supertest npm install --save-dev cypress

```
Administrator: Command Prompt
C:\Windows\System32\testing-darcy>npm install --save-dev jest
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if
if you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerfu
1.
npm warn deprecated glob@7.2.3: Glob versions prior to v9 are no longer supported
added 298 packages, and audited 299 packages in 2m
44 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities

C:\Windows\System32\testing-darcy>npm install --save-dev mocha chai
added 30 packages, and audited 329 packages in 9s
54 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities

C:\Windows\System32\testing-darcy>npm install --save-dev supertest
added 39 packages, and audited 368 packages in 15s
69 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities

C:\Windows\System32\testing-darcy>npm install --save-dev cypress
added 132 packages, and audited 500 packages in 30s
85 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
```

Code quality tools

npm install --save-dev eslint npm install --save-dev nyc

Development dependencies npm install express

```
Administrator: Command Prompt

C:\Windows\System32\testing-darcy>npm install --save-dev eslint

added 67 packages, and audited 567 packages in 20s

99 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Windows\System32\testing-darcy>npm install --save-dev nyc
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported
npm warn deprecated glob@7.2.3: Glob versions prior to v9 are no longer supported
npm warn deprecated glob@7.2.3: Glob versions prior to v9 are no longer supported

added 54 packages, and audited 621 packages in 9s

105 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Windows\System32\testing-darcy>npm install express

added 47 packages, and audited 668 packages in 12s

107 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

Step 5: Configuration Files

Create configuration files:

package.json scripts section:

```
{ package.json > type
1  {
2    "name": "testing-darcy",
3    "version": "1.0.0",
4    "type": "module",
5    "main": "index.js",
6    "scripts": {
7      "start": "node app.js",
8      "test": "jest",
9      "test:mocha": "mocha",
10     "test:coverage": "nyc jest",
11     "test:coverage:all": "npm run test:coverage && npm run test:coverage:mocha",
12     "lint": "eslint .",
13     "lint:fix": "eslint . --fix",
14     "cypress:open": "cypress open"
15   },
```

eslint.config.js (ESLint v9+ configuration):

```

eslint.config.js > [⌕] default
1  import js from '@eslint/js';
2
3  export default [
4    js.configs.recommended,
5    {
6      languageOptions: {
7        globals: {
8          describe: 'readonly',
9          test: 'readonly',
10         expect: 'readonly',
11         beforeEach: 'readonly',
12         afterEach: 'readonly',
13         beforeAll: 'readonly',
14         afterAll: 'readonly'
15       }
16     }
17   ]
18 ];

```

jest.config.js:

```

JS jest.config.js > [⌕] default
1  export default {
2    testEnvironment: "node",
3    testMatch: ["**/tests/jest/**/*.test.js"],
4    collectCoverage: true,
5    coverageDirectory: "coverage",
6    coverageReporters: ["text", "lcov", "html"],
7    transform: {},
8    testPathIgnorePatterns: ["/node_modules/", "/tests/mocha/"],
9  };
10

```

Step 6: Create Verification Tests

[basic.test.js](#):

Jest test

```

JS basic.test.js > ...
1 describe('Environment Setup Vervicarion',()=>{
2     test('Node.js is working' ,() =>{
3         expect (typeof process.version).toBe('string');
4     });
5     test('Basic math operaions',()=>{
6         expect(2+2).toBe(4);
7     });
8     test ('ES modules are working',()=>{
9
10
11
12         const testFunction = () => 'ES modules work!';
13
14
15         expect(testFunction()).toBe('ES modules work!');
16
17     });
18
19 });

```

app.js

```

JS app.js > app.listen() callback
1 import express from 'express';
2
3 const app = express();
4 const PORT = process.env.PORT || 3000;
5
6 app.use(express.json());
7
8 app.get('/', (req, res) => {
9     res.json({
10         message: 'Testing environment ready!',
11         moduleType: 'ES modules',
12         nodeVersion: process.version
13     });
14 });
15
16 app.get('/health', (req, res) => {
17     res.json({
18         status: 'healthy',
19         timestamp: new Date().toISOString()
20     });
21 });
22
23 app.listen(PORT, () => {
24     console.log('Server running on port ${3000}');
25 });
26
27 export default app;

```

api.test.js

```
JS api.test.js > ...
1  import request from 'supertest';
2  import app from '../app.js';
3  import { describe, it } from 'mocha';
4  import { expect } from 'chai';
5
6  describe('API Tests with ES Modules', () => {
7    it('GET / should return success message', async () => {
8      const response = await request(app).get('/');
9      expect(response.status).to.equal(200);
10     expect(response.body.message).to.equal('Testing environment ready!');
11     expect(response.body.moduleType).to.equal('ES modules');
12   });
13
14   it('GET /health should return health status', async () => {
15     const response = await request(app).get('/health');
16     expect(response.status).to.equal(200);
17     expect(response.body.status).to.equal('healthy');
18     expect(response.body.timestamp).to.not.be.undefined;
19   });
20 });
```

math.js

```
JS math.js > ...
1  const { add, multiply, divide } = require('../math.js');
2  export const add = (a, b) => a + b;
3  export const multiply = (a, b) => a * b;
4  export const divide = (a, b) => {
5    if (b === 0) throw new Error('Division by zero');
6    return a / b;
7  };
8
9  export default {
10    add,
11    multiply,
12    divide
13  };
```

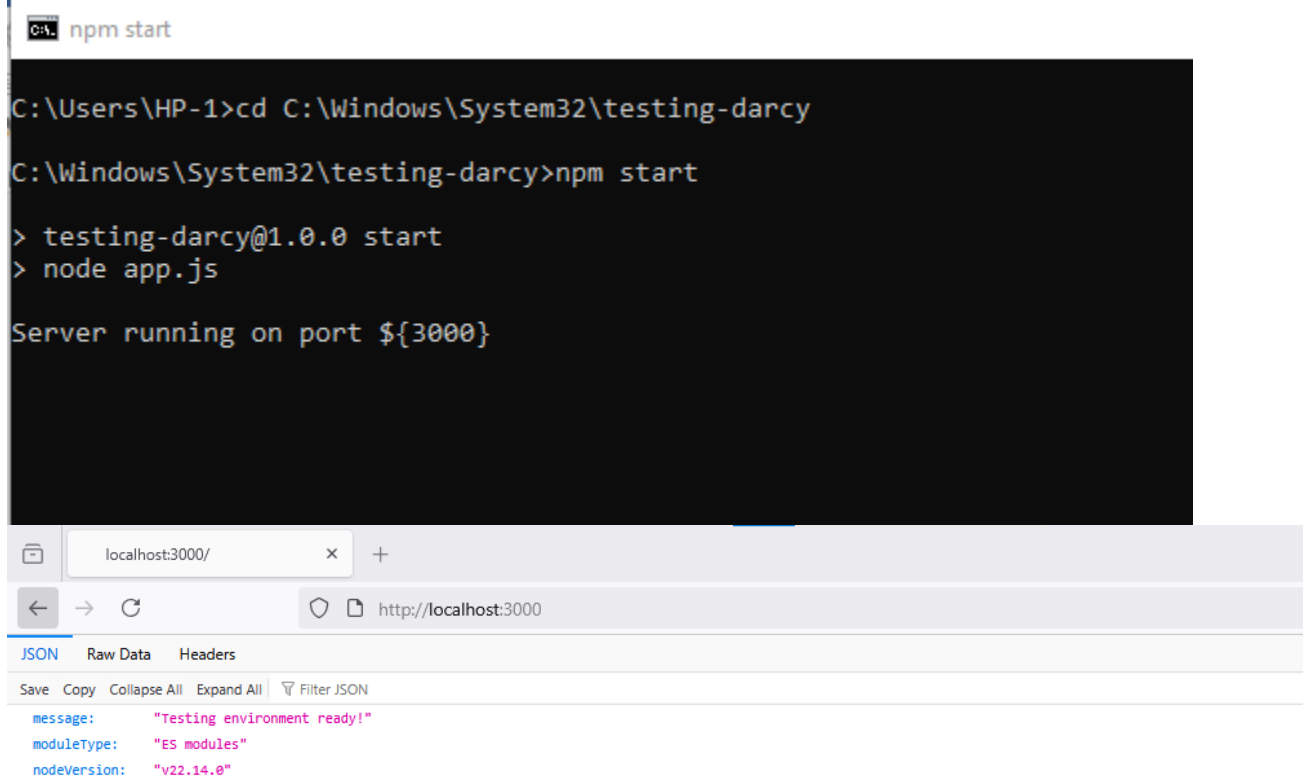

math.test.js

```
JS math.test.js > ...
1  const { add, multiply, divide } = require('.././math.js');
2
3  import mathModule from '.././math.js';
4  import { describe, it } from 'mocha';
5  import { expect } from 'chai';
6
7  describe('Math Module Tests', () => {
8    it('add function works correctly', () => {
9      expect(add(2, 3)).to.equal(5);
10     expect(add(-1, 1)).to.equal(0);
11   });
12
13   it('multiply function works correctly', () => {
14     expect(multiply(3, 4)).to.equal(12);
15     expect(multiply(-2, 3)).to.equal(-6);
16   });
17
18   it('divide function works correctly', () => {
19     expect(divide(10, 2)).to.equal(5);
20     expect(() => divide(10, 0)).to.throw('Division by zero');
21   });
22
23   it('default export works', () => {
24     expect(mathModule.add(5, 5)).to.equal(10);
25   });
26 });
```

.mocharc.json

```
{ } .mocharc.json > ...
1  {
2    "spec": "tests/mocha/*.js",
3    "recursive": true,
4    "timeout": 5000
5  }
```

Step 7: Run Verification Tests , npm start



The screenshot shows a terminal window with the following commands and output:

```
C:\Users\HP-1>cd C:\Windows\System32\testing-darcy
C:\Windows\System32\testing-darcy>npm start

> testing-darcy@1.0.0 start
> node app.js

Server running on port ${3000}
```

Below the terminal window, a web browser is open to <http://localhost:3000>. The browser's developer tools are open to the 'JSON' tab, displaying the following response:

```
{
  "message": "Testing environment ready!",
  "moduleType": "ES modules",
  "nodeVersion": "v22.14.0"
}
```

Run tests npm test
npm run test:mocha

```
S C:\Windows\System32\testing-darcy> npm run
testing-darcy@1.0.0 test:mocha
mocha
```

server running on port 3000

API Tests with ES modules

GET / should return success message(46ms)

GET /health should return health status

Math Module Tests

- ✓ add function works correctly
- ✓ multiply function works corretly
- ✓ divide function works correctly
- ✓ default export works

passing (97ms)

npm run test:cov

```

PS C:\Windows\System32\testing-darcy> npm run
test:coverage
> testing-darcy10.0 test:coverage
> nyc
PASS tests/basic.test.js
✓ Environment Setup Veriification
✓ Jest Node is working (ms)
✓ Basic math operations (ms)
✓ ES modules are working (ms)
+-----+-----+-----+-----+
| % Stmts   | % Branc | % Lins | Uncovered Line #s |
+-----+-----+-----+-----+
| tests/basic.|    100  |    100  |                    0 |
Test Suites: 1 passed, 1 total
Tests: 3 passed, 3 total
Time: 0.755 s, estimated
+-----+-----+-----+-----+
| Test Suites | Total | Uncovered Line #s |
+-----+-----+-----+-----+
| tests/basic.test.js |    100  |                    100 |

```

npm run test:coverage:mocha

```
> testing-shema@1.0.0 test:coverage:mocha  
> nyc mocha
```

Server running on port 3000

API Tests with ES Modules

- ✓GET / should return success message
- ✓GET /health should return health status

Math Module Tests

- ✓add function works correctly
- ✓multiply function works correctly
- ✓divide function works correctly
- ✓default export works

6 passing (39ms)

npm run test:coverage:all

```
PS C:\Windows\System32\testing-darcy> npm run
test:coverage && npm run test:coverageall
> nyc jest
```

```
PASS tests/basic.test.js
```

- ✓ Environment Setup Verification
- ✓ Jest Node is working
- ✓ Basic math operations works

=====	% \$stms	% Branchd	% Funcs	Unc=====
File	=====	=====	=====	iredLine #s
=====	100 %	100 %	100%	=====
all	-----	-----	-----	100 %
+-----+	---100 %	-----	-----	+-----+

```
Test Suites: 1 passed, 1 total
```

```
Ran all test suites 1 s
```

```
testing-darcy@1.0 test:coverage:moha
```

```
> nyc mocha
```

```
API Tessts with ES Modules
```

- ✓ GET /should return a success message(46ms)
- ✓ GET /should return health status

```
Math Module Tests
```

- ✓ add function works correctly
- ✓ multiply function works correctly
- ✓ default export works

```
6 passing (259ms)
```

```
PS C:\Windows\Sytem32\testing-darcy>
```

npm run lint

```
PS C:\Windows\System32\testing-darcy> npm run lint

> testing-darcy@1.0.0 lint
> eslint .

C:\Windows\System32\testing-darcy\app.js
   4:14  error  'process' is not defined  no-undef
  12:18  error  'process' is not defined  no-undef
  24:3   error  'console' is not defined  no-undef

C:\Windows\System32\testing-darcy\basic.test.js
   3:24  error  'process' is not defined  no-undef

C:\Windows\System32\testing-darcy\math.js
   2:14  error  Parsing error: Identifier 'add' has already been declared

C:\Windows\System32\testing-darcy\math.test.js
   1:35  error  'require' is not defined  no-undef

✖ 6 problems (6 errors, 0 warnings)

PS C:\Windows\System32\testing-darcy> 
```

npm run cypress:open

```
PS C:\Windows\System32\testing-darcy> npm run cypress:open

> testing-darcy@1.0.0 cypress:open
> cypress open

DevTools listening on ws://127.0.0.1:64701/devtools/browser/4c456630-db5f-47de-9f59-80095230b15e
> 
```




Welcome to Cypress!

[Review the differences between each testing type →](#)



E2E Testing

Build and test the entire experience of your application from end-to-end to ensure each flow matches your expectations.

Not Configured



Component Testing

Build and test your components from your design system in isolation in order to ensure each state matches your expectations.

Not Configured

